21-242 Recitation 12

Linear Transformations — Q5. Surjectivity iff the image of a basis spans

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Let $\mathcal{B} \subseteq V$ be a basis. Prove that $T : V \to W$ is surjective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ spans W.

Course-level solution

Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_n)$. For any $\vec{v} = \sum a_i \vec{b}_i$,

$$T(\vec{v}) = \sum a_i T(\vec{b}_i),$$

so

$$\operatorname{im}(T) = \operatorname{Span}\big(T(\vec{b}_1), \ldots, T(\vec{b}_n)\big).$$

Therefore T is surjective iff $\operatorname{im}(T) = W$ iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ spans W.